

## Business Analyst's Guide to Integration Touchpoints in Procurement Systems

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### Purpose of This Guide

This guide is designed to help new Business Analysts understand and document **integration touchpoints** within the context of enterprise systems, specifically focused on **Procurement Management Portals (PMP)** and systems like **SAP, Tally, Zoho**, and vendor/payment platforms. It is structured to explain concepts from the ground up, just like a mentor would.

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### Part 1: What Are Integration Touchpoints?

**Definition:** Integration touchpoints are the moments where **two or more systems** exchange **data, commands, or status updates** to keep business workflows connected and automated.

**Examples:** - PMP portal sends PO data to SAP - GRN generated in PMP updates stock in Inventory tool - Payment status from Tally pushed back to PMP

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### Part 2: Why Do They Matter for a BA?

As a BA, you are responsible for: - Understanding where systems interact - Defining what data flows between them - Documenting how those flows are triggered and validated - Communicating integration needs to technical teams (devs, architects)

You will work closely with **system architects, developers, QA, and external vendors**.

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### Part 3: Common Procurement Integration Touchpoints

| # | Integration Point   | Systems Involved               | Data Passed                      | Trigger Event          | Type (Push / Pull / Sync) |
|---|---------------------|--------------------------------|----------------------------------|------------------------|---------------------------|
| 1 | Request Data Export | PMP → ERP (SAP/ Tally)         | Request ID, Items, Qty           | Request Approved       | Push                      |
| 2 | PO Generation       | PMP → SAP / Tally              | PO ID, Vendor ID, Items, Rate    | After Vendor Selection | Push                      |
| 3 | GRN Sync            | Warehouse Tool → PMP / SAP     | GRN ID, PO ID, Quantity Received | Item Received          | Push / Sync               |
| 4 | Invoice Upload      | Vendor Portal → PMP / Accounts | Invoice Number, PO, GRN, Amount  | Vendor Upload          | Push                      |
| 5 | 3-Way Matching      | PMP + SAP + Tally              | PO, GRN, Invoice                 | Upon Invoice Upload    | Pull / Sync               |

| # | Integration Point | Systems Involved   | Data Passed                  | Trigger Event     | Type (Push / Pull / Sync) |
|---|-------------------|--------------------|------------------------------|-------------------|---------------------------|
| 6 | Payment Status    | Tally / Bank → PMP | Paid/Unpaid + Reference ID   | Payment Processed | Push                      |
| 7 | Status Update     | SAP / PMP → Vendor | PO Status, Delivery Expected | PO Issued / GRN   | Push                      |

## Part 4: Types of Integration

| Type                    | Description                                | Example                                      |
|-------------------------|--------------------------------------------|----------------------------------------------|
| <b>Real-Time API</b>    | Instant data exchange via web APIs         | PMP calls SAP API to create PO               |
| <b>Batch Upload</b>     | Data synced in bulk at scheduled intervals | Nightly sync of GRNs from warehouse system   |
| <b>Manual Upload</b>    | User exports/imports CSV files             | Vendor uploads invoice manually              |
| <b>Webhooks</b>         | Automated callback after event             | Tally webhook notifies PMP of payment status |
| <b>Middleware (ESB)</b> | Integration via central bus or service     | SAP, PMP, and Tally communicate via MuleSoft |

## Part 5: BA Tasks for Integration Documentation

| Task                         | Description                                                     |
|------------------------------|-----------------------------------------------------------------|
| 1. Identify Systems          | Understand all systems involved (PMP, ERP, Finance, Vendor)     |
| 2. Document Data Fields      | What fields are exchanged (e.g., PO ID, amount, GRN ID)         |
| 3. Create Integration Matrix | Table with source, destination, trigger, frequency              |
| 4. Define Error Scenarios    | What happens if data doesn't flow or fails validation?          |
| 5. Collaborate with Devs     | Finalize API specs, formats (JSON/XML), retry logic             |
| 6. Support Testing           | Help QA design test cases, mock payloads, and simulate failures |

## Part 6: Sample Integration Mapping Template

| # | Source System | Destination System | Field List         | Trigger     | Method | Frequency |
|---|---------------|--------------------|--------------------|-------------|--------|-----------|
| 1 | PMP           | SAP                | PO ID, Items, Rate | On Approval | API    | Real-time |

| # | Source System | Destination System | Field List             | Trigger     | Method  | Frequency |
|---|---------------|--------------------|------------------------|-------------|---------|-----------|
| 2 | Vendor        | PMP                | Invoice No, Amount     | File Upload | Manual  | On Demand |
| 3 | Tally         | PMP                | Payment Status, Ref No | On Payment  | Webhook | Real-time |

## Common Issues & How to Handle Them

| Issue             | Root Cause                          | BA Solution                           |
|-------------------|-------------------------------------|---------------------------------------|
| Missing Data      | Source field is null                | Add mandatory field checks            |
| Delay in Sync     | Batch runs once daily               | Schedule changes / alert users        |
| Format Mismatch   | Field type mismatch (int vs string) | Add field format rules in mapping doc |
| Duplicate Records | No uniqueness constraints           | Add UUID or composite keys            |
| Partial Updates   | System crash during sync            | Design retry + rollback mechanisms    |

## Tools You Can Use

- **Draw.io / Lucidchart:** For system diagrams and data flow maps
- **Excel:** For integration matrix and field mapping
- **Swagger / Postman:** To test and view API specs
- **JIRA / Confluence:** To track integration-related stories, tasks

## Final Words

As a BA, your job is to make integration "invisible" to the user — but rock solid in the backend. You don't need to code APIs, but you must know **what goes in, what comes out, and what can go wrong.**

Every integration you define improves automation, efficiency, and user satisfaction. You become the translator between systems, teams, and business needs.

Welcome to the system integration layer — it's where real transformation happens!